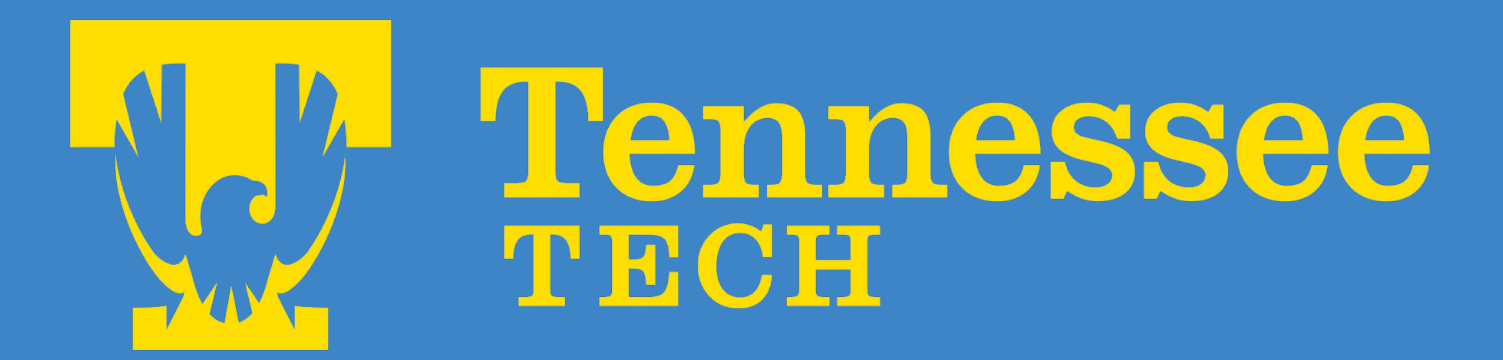




Early-Life Production Forecasting for Unconventional Wells Using XGBoost

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INTRODUCTION

Problem Statement

Decline Curve Analysis (DCA) is the industry standard for oil production forecasting and performs well once a well has 12–18 months of production history and enters stable decline. For unconventional horizontal wells, operators must decide to allocate capital for new wells far earlier, often with just an early time threshold of data, before the decline trend is established. DCA's early time fits are unreliable in these conditions, producing large forecast errors that drive poor investment decisions.

Objective

Develop a machine learning model that forecasts monthly oil production out five years using an early time threshold of a well's production history, and demonstrate that it materially outperforms the DCA early time baseline on independent, unseen wells.

METHODOLOGY

Decline Curve Analysis (DCA)

Each well's production history was fit the Arps Hyperbolic Decline Curve model, which characterizes production rate over time using three parameters: baseline peak rate, baseline curve shape, and the baseline decline coefficient. Fitting was performed using non-linear least squared optimization via the Levenberg-Marquardt algorithm, with physically grounded parameter bounds applies to prevent nonsensical forecasts. A early time threshold fit using only a limited set of monthly data was applied to establish a baseline empirical forecast and evaluate accuracy against remaining production history.

XGBoost

An XGBoost regressor was trained to forecast monthly oil production rates from 5 years out from the early time threshold of a well's production history, the same early-life constraint faced by DCA. The 18-feature input set included individual monthly production rates, derived statistics, the three Arps parameters from the early time DCA fit, well surface coordinates, and time polynomial features. Production targets were log1p-transformed to address the right-skewed rate distribution across the well population. To measure model interpretability, we used SHAP values, seen in Figure 4.

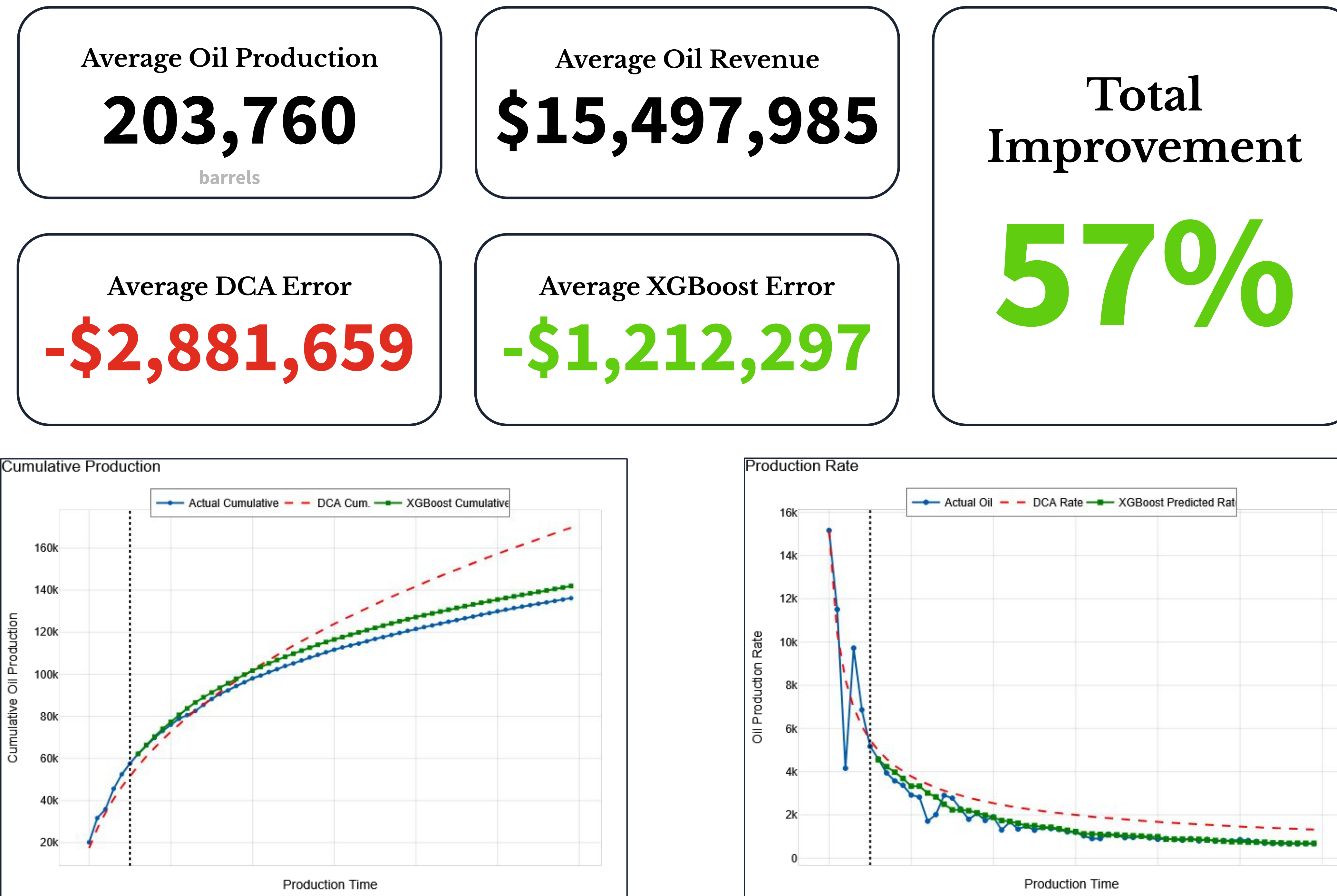


Figure 1. Cumulative production for a well from the unseen test set

Figure 2. Monthly production rate for a well from the unseen test set

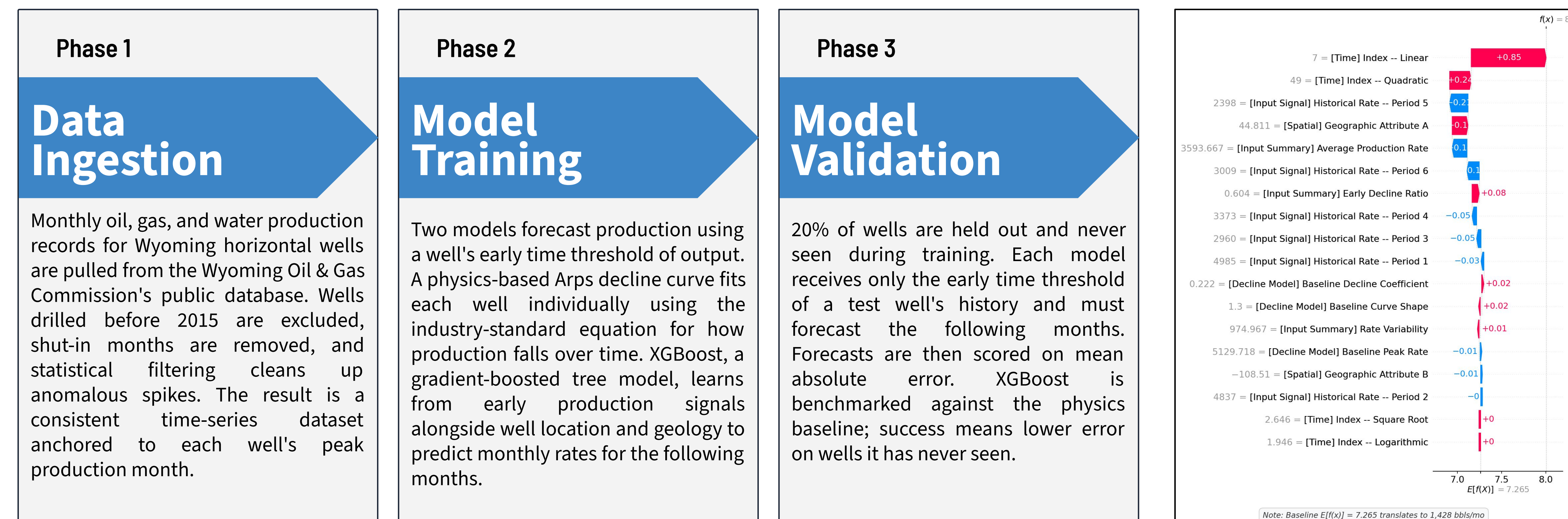


Figure 3. Overview of the three-phase machine learning pipeline

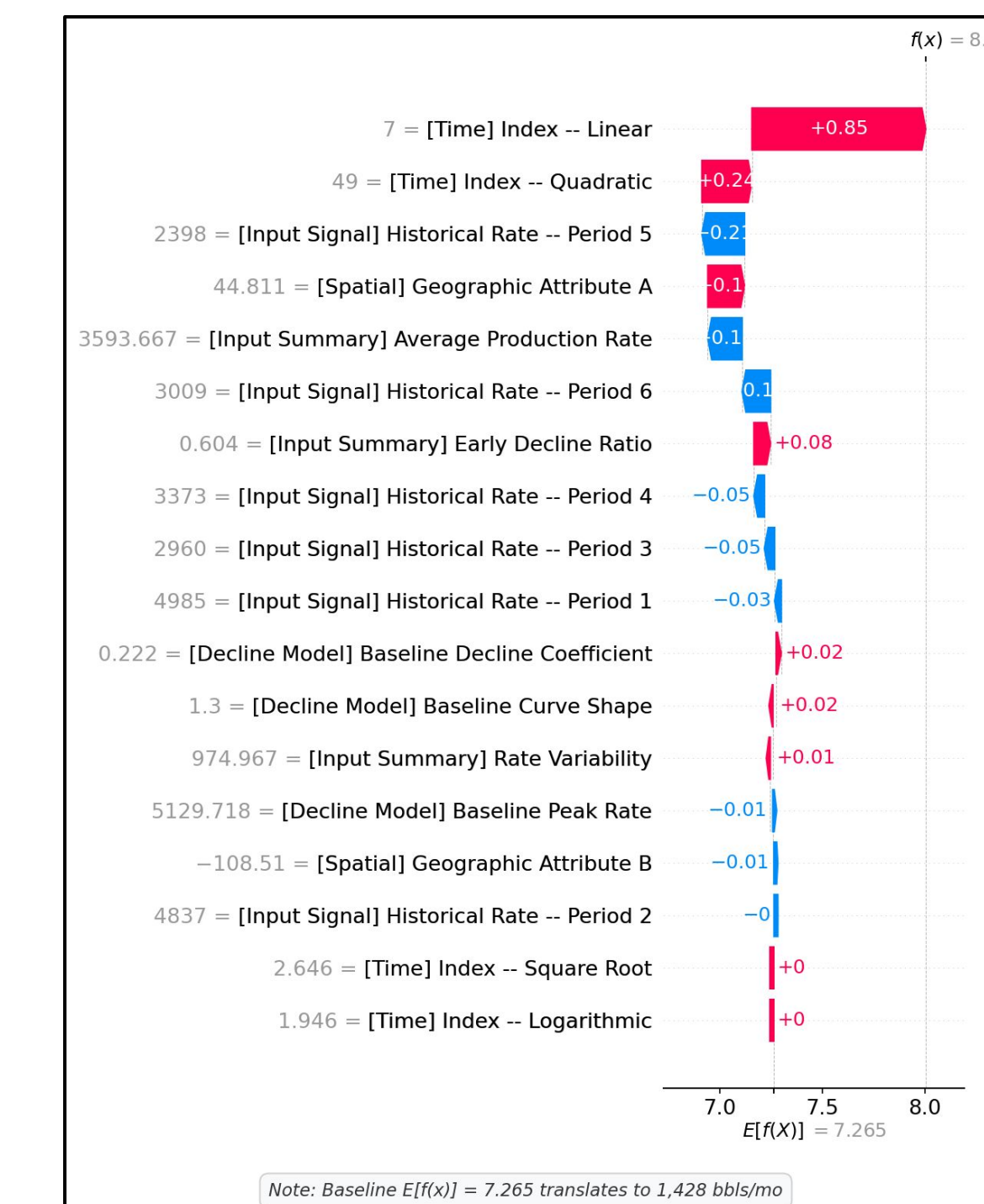


Figure 4. Shapley feature importance plot

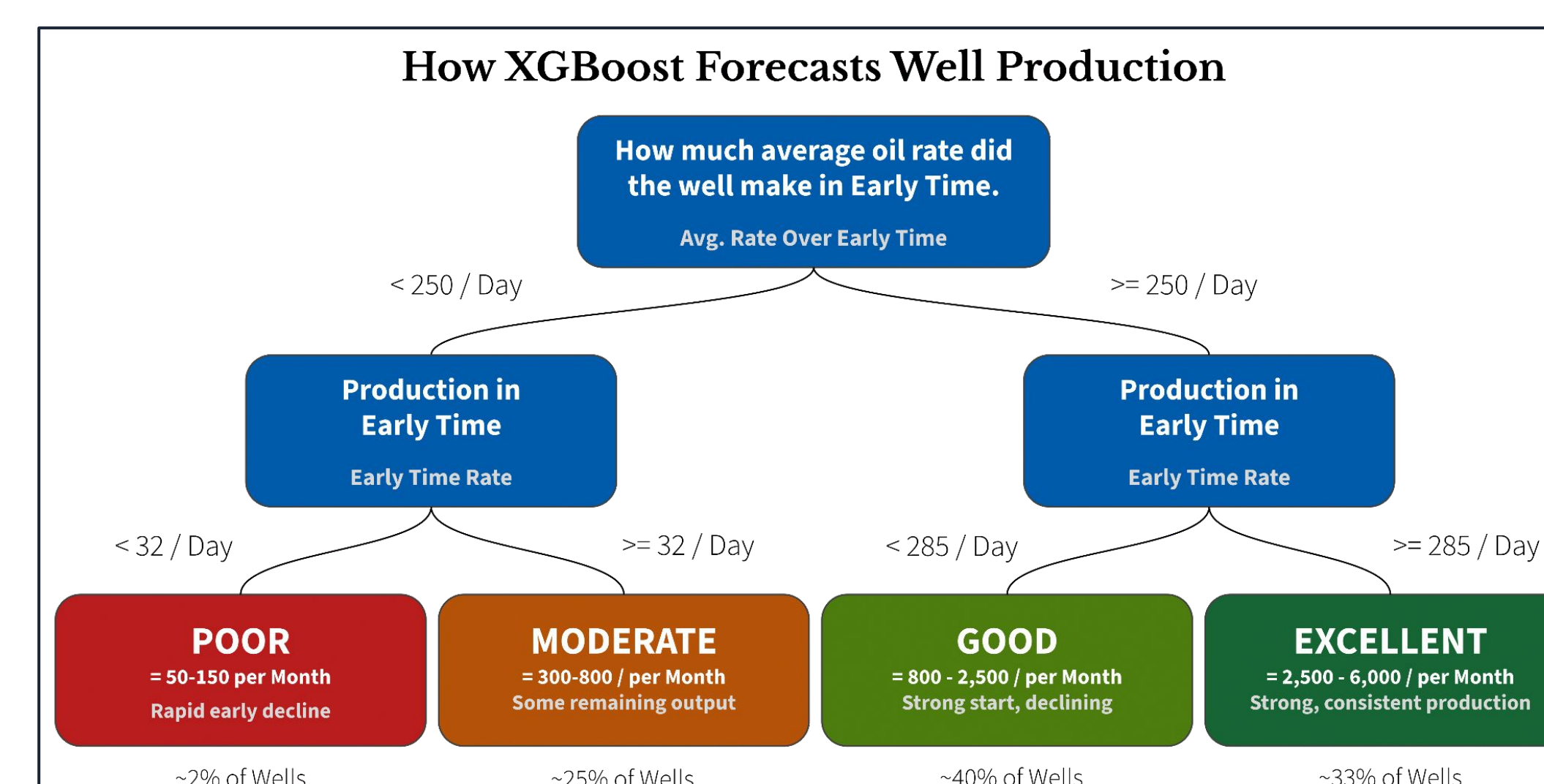


Figure 5. A simplified representation of the XGBoost model

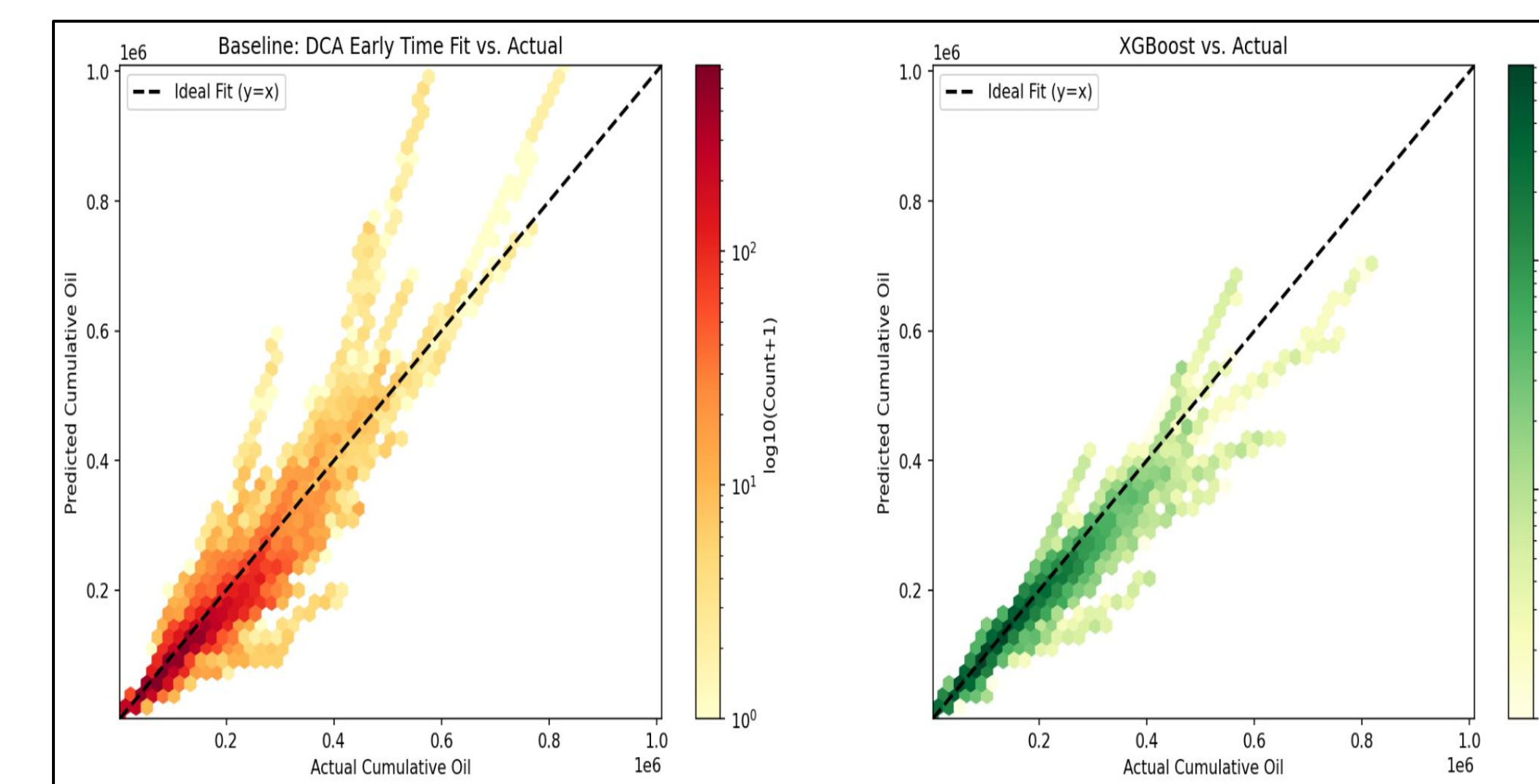


Figure 6. Predicted vs. actual cumulative oil production across all test wells, with color indicating well density.

RESULTS

Across the test set, XGBoost achieved an average forecast error of 15,938 bbl per well compared to 37,938 bbls for the DCA early time threshold, a 57% reduction in MAE. The financial impact is equally significant: DCA's average revenue projection error was -\$2,881,659 per well, while XGBoost reduced this to -\$1,212,297. The SHAP Feature Importance (Figure 4) visualizes the logic behind these forecasts, with early production signals and well location emerging as the strongest predictors of long-term output.

By reducing forecast "overshoot," XGBoost protects capital and ensures new well placement decisions are backed by a reliable revenue floor rather than a risky projection. Figure 6 confirms strong model accuracy across the full test set, with predicted cumulative production tracking closely to actual values across all unseen wells.

DISCUSSION

With our model consistently outperforming the baseline, industry leaders can make informed decisions on where to place new wells. When a single well in a field performs well, it is a strong indicator of a similar well's performance in that area, leading to expensive new wells to be dug. If they discover in five years that they produce only half of the DCA projection, they may realize significantly lower returns.

Traditional DCA projections risk over-leveraging on wells that may underperform years later. In contrast, XGBoost's strategic value lies in its conservative bias. By significantly reducing "overshoot" compared to the industry baseline, it protects capital and ensures that new well placement is backed by a reliable revenue floor rather than a risky bet.

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